Achouri Youcef Islem

Machine Learning Research Engineering Internship

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Final-year ESI-SBA engineering student interested in speech representations, sequence models and reinforcement learning. Available January 2027; seeking an internship with careful experimentation and evaluation.

EDUCATION

ESI-SBA 2021 – 2027 (expected)
École Nationale Supérieure d'Informatique de Sidi Bel Abbès
State Engineering Degree in Computer Science (Bac+5) · Artificial Intelligence and Data Structures

PROFESSIONAL EXPERIENCE

Partner and Lead Spatial Systems Engineer March 2026 – Present
Azurid · AzuridTwinUp case study

- Built AzuridTwinUp, an authenticated industrial digital twin platform, with a Three.js/React viewer and NestJS/Prisma backend. Recorded viewer demonstration available.
- Worked across Matterport Pro3 LiDAR and DJI Mini 3 image acquisition, alignment troubleshooting in Metashape/RealityScan, E57 processing, and Meshopt/KTX2 delivery.

SELECTED PROJECTS

StethoSpeech research adaptation 2026
PyTorch · HuBERT · Transformers

- Adapted Shah et al. (IMWUT 2024): trained a 19.7M-parameter Transformer from murmur features to HuBERT representations of TTS-generated speech.
- Recorded 0.23171 feature MAE on 1,133 held-out utterances from familiar speakers; trained 45 epochs, selecting epoch 35. Speech intelligibility was not evaluated.

Deep Q-learning experiments 2026
PyTorch · Gymnasium

- Implemented DQN, Double DQN and dueling networks with replay buffers, target updates and greedy evaluation across control, Atari and custom grid environments.

English–French translation architectures 2026
PyTorch · Attention · Transformers

- Implemented BiLSTM, Bahdanau attention, decoder-only and encoder–decoder Transformers, with attention masks and positional encodings.

Neural network from scratch 2026
Python · NumPy

- Implemented a bias-free 784–128–10 MLP and analytical backpropagation. Public MNIST baseline: 91.62% ± 0.23 percentage points across five seeds, on 9,984 test images.

TECHNICAL SKILLS

Research: Python, PyTorch, NumPy, HuBERT, Transformers, Gymnasium.
Methods and tools: Backpropagation, attention, experimental evaluation, Git, Linux, pytest.

LEADERSHIP AND AWARDS

Vice President, INGENIUMS Club 2023 – 2024

- Co-organized technical and cultural events; invited artists and regional student delegations, coordinated teams, and presented projects to visitors and school leadership.

Awards: 2nd place, DevFest Hackathon 2023 · 3rd place, NCS Hack 3.0, June 2026.
Languages: Arabic (native), English (fluent), French (fluent).